

Phillip Liems

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Engineering leader with progressive experience in rail vehicle design, system architecture, production support, project controls, and manufacturing engineering. Skilled in leading global engineering teams, developing technical solutions for rail vehicle platforms, supporting bids and customer requirements, and driving design integration from concept through production. Strong background in interior systems, customer interfaces, supplier coordination, schedule management, and manufacturability.

Core Skills

System Architecture | Rail Vehicle Design | Technical Leadership | Interior & Customer Interface Systems | Production Support | Bid & Proposal Support | Design-to-Cost | Supplier Coordination | Project Controls | Capacity Planning | Lean / Six Sigma | CREO / ProE | 3D Printing & Prototyping | Cross-Functional Team Leadership

Professional Experience

Siemens Mobility — Sacramento, CA

System Architect | Feb 2026 – Present

- Lead a global team of design engineers delivering technical solutions for current production rail vehicle programs.
- Own system architecture, technical design, and integration of interior and customer interface systems for Coach, Light Rail, and Locomotive vehicles.
- Support new designs and bid proposals by defining technical concepts, customer requirements, and initial cost estimates in collaboration with procurement.
- Provide technical leadership from concept through production support, ensuring solutions meet project, customer, operational, and integration requirements.

Component Engineer | Feb 2019 – Jan 2026

- Designed and released multiple installation packages across various systems, coach types, and rail vehicle contracts.
- Consistently met or exceeded design schedule commitments while supporting team deliverables and production needs.
- Collaborated with suppliers from First Article Inspection through series production to support manufacturability and quality.
- Supported onsite production by resolving manufacturing issues, identifying root cause, and providing redlines and repair procedures.
- Created hundreds of parts, assemblies, weldments, and installations currently in revenue service; accumulated 10,000+ hours of ProE/CREO experience.
- Led training on advanced design techniques, model robustness, and 3D printing for prototyping and fit-checking engineering solutions.

Project Controls Engineer | Sep 2016 – Feb 2019

- Analyzed customer contractual requirements and developed internal and customer-facing schedules to support manufacturing execution.
- Partnered with Operations Directors to align production sequence, takt requirements, inspections, and downstream process needs.
- Evaluated historical data, supplier risks, material availability, lead times, workload, and headcount requirements to forecast production impacts.
- Identified bottlenecks and recommended resource needs, including manpower, fixtures, equipment, storage, and manufacturing space.

Designer / Drafter II | Nov 2015 – Sep 2016

- Designed and released Light Rail Vehicle carshell components aligned with customer requirements and structural calculation inputs.
- Created piece parts, assembly models, weldment drawings, inspection criteria, part numbers, and bills of material.
- Built a dimensionally accurate train interior mockup to support customer review of ADA compliance requirements.

Industrial Engineer / Special Assignment: Carshell Champion | Sep 2014 – Nov 2015

- Performed capacity planning for new designs to ensure required space, equipment and headcount to support delivery requirements.
- Developed fixtures, jigs, procedures, and standard work instructions using Lean and Six Sigma principles to improve safety, quality, and efficiency.
- Coordinated engineering, purchasing, production, receiving, and supplier activities to manage an alternate carshell material process and maintain delivery schedules.

Education

Bachelor of Science, Mechanical Engineering

California State University, Sacramento | May 2014

Dean's List, Junior and Senior Year | Tau Beta Pi Engineering Honor Society | Eagle Scout, Troop 87, Napa, CA